Hoor Mohamed Magdy Mahmoud

AI & Machine Learning Engineer | Software Developer

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LinkedIn | GitHub

Professional Summary

AI/ML engineer and software developer with a Computer Science degree (Very Good) from MTI. Experienced in building, training, and evaluating machine learning models, from classical algorithms to deep learning (CNN, RNN, LSTM, GAN), and in delivering e-commerce and software solutions for clients as a remote developer. Completed hands-on projects in forecasting, computer vision, and recommendation systems, plus formal training in AI (ITI) and Full Stack .NET development (DEPI). Fast learner who turns problems into working, usable solutions.

Technical Skills

- **Programming Languages:** Python, C#, C++, JavaScript/TypeScript, MATLAB
- **AI & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, Model Training and Evaluation, Deep Learning (CNN, RNN, LSTM, GAN), Computer Vision, Recommendation Systems
- **Web Development:** ASP.NET, HTML5, CSS3, JavaScript/TypeScript, PHP (Full Stack Development)
- **E-commerce Platforms:** Salla, Zid, Father Shop, Shopify
- **Frameworks and Tools:** .NET Framework (Windows Forms), Streamlit, Jupyter, Git
- **Databases:** SQL Server, MySQL
- **Office Tools:** Microsoft Office (Advanced)

Professional Experience

Software Developer (Remote) | AD FAZ, Saudi Arabia | September 2025 – 2026

- Built e-commerce stores for clients on Salla, Zid, Father Shop, and Shopify.
- Developed and maintained software solutions for client projects.
- Collaborated with cross-functional teams to deliver scalable, production-ready applications.

Machine Learning Developer | Prodigy InfoTech | 2024

- Developed and optimized machine learning models for real-world applications.

Machine Learning Developer | CodeAlpha | 2024

- Contributed to AI-driven projects focused on data analysis and predictive modeling.

Projects

Stock Price Prediction: Investigating Practical Models and Techniques (Graduation Project) | September 2023 – May 2024

- Built and compared 11 ML and deep learning models for stock price forecasting: Simple and Multiple Linear Regression, SVR, Decision Tree, Random Forest, KNN, MLP, RNN, CNN, GAN, and LSTM.
- Developed a user-friendly Streamlit interface that presents investment insights to end users.
- Tools: Python, TensorFlow, PyTorch, Jupyter Notebook, Anaconda.

DEPI TOTC: Learning Management System (DEPI Graduation Project) | GitHub

- Built a full-stack Learning Management System as a team for the DEPI graduation project, with 5 user roles (SuperAdmin, Admin, Instructor, Student, Parent), each with a tailored dashboard.
- AI Exam Generator: instructors upload a PDF and Mistral AI generates structured MCQ and True/False questions from its content.
- RAG-based Student Chatbot: students upload study materials and get context-aware answers.
- Weekly schedule management with Zoom integration, conflict detection, and role-filtered views.
- Also includes a blog and news system, automated weekly parent reports via n8n, support ticketing, and announcements.
- Tools: Mistral AI, RAG, n8n, Zoom integration.

Food Recognition and Calorie Estimation System | 2024 | GitHub

- Built a model that recognizes food items from images and estimates their calorie content.
- Helps users track their dietary intake through AI-powered food analysis.

Hand Gesture Recognition System | 2024 | GitHub

- Developed a model that identifies and classifies hand gestures from image and video data.
- Enables intuitive gesture-based control for interactive applications.

Cat vs. Dog Image Classification with SVM | 2023 | GitHub

- Implemented a Support Vector Machine classifier for binary image classification on a Kaggle dataset.
- Improved accuracy through image augmentation and hyperparameter tuning.

Music Recommendation System | 2023 | GitHub

- Built a recommendation engine combining collaborative filtering and content-based approaches.
- Improved the user experience with personalized music suggestions.

Customer Segmentation with K-Means Clustering | 2023 | GitHub

- Developed a K-Means clustering solution to segment retail customers by purchase history.
- Supported targeted marketing strategies through customer behavior analysis.

House Price Prediction with Linear Regression | 2023 | GitHub

- Built a linear regression model predicting house prices from square footage, bedrooms, and bathrooms.
- Improved prediction accuracy through feature engineering and data preprocessing.

Education

Bachelor's Degree in Computer Science | Modern University for Technology and Information (MTI) | October 2020 – July 2024

- Grade: Very Good
- Language of instruction: English

Certifications and Courses

- **Software Development: Full Stack .NET Web Developer**, Digital Egypt Pioneers Initiative (DEPI) , 2026
- **Artificial Intelligence**, Information Technology Institute (ITI), 2023
- Multiple specialized courses in AI, Machine Learning, and Data Science, 2023–2024

Languages

- **Arabic:** Native
- **English:** Very Good